

Glorot *et al.*, 2011), whose non-linearity is simply $\max(0, \arg)$, i.e.,

$$h_i = \max(0, z_i + a_i)$$

where z_i is a zero-mean noise. Although in practice Gaussian noise is appropriate, interesting properties can be shown for z_i sampled from the logistic density $p(z) = \text{sigm}(z)(1 - \text{sigm}(z))$, where $\text{sigm}(\cdot)$ represents the logistic sigmoid function.

In that case, we can prove nice properties of the noisy rectifier.

Proposition 1. *The noisy rectifier with logistic noise has the following properties: (a) $P(h_i > 0) = \text{sigm}(a_i)$, (b) $E[h_i] = s_+(a_i)$, where $s_+(x) = \log(1 + \exp(x))$ is the softplus function and the random variable is z_i (given a fixed a_i).*

Proof.

$$\begin{aligned} P(h_i > 0) &= P(z_i > -a_i) = 1 - \text{sigm}(-a_i) \\ &= 1 - \frac{1}{1 + e^{a_i}} = \frac{1}{1 + e^{-a_i}} = \text{sigm}(a_i) \end{aligned}$$

which proves (a). For (b), we use the facts $\frac{d \text{sigm}(x)}{dx} = \text{sigm}(x)(1 - \text{sigm}(x))$, $s_+(u) \rightarrow u$ as $u \rightarrow \infty$, change of variable $x = a_i + z_i$ and integrate by parts:

$$\begin{aligned} E[h_i] &= \lim_{t \rightarrow \infty} \int_{z_i > -a_i}^t (a_i + z_i) \text{sigm}(z_i) (1 - \text{sigm}(z_i)) dz_i \\ &= \lim_{t \rightarrow \infty} \left([x \text{sigm}(x - a_i)]_0^t - \int_0^t \text{sigm}(x - a_i) dx \right) \\ &= \lim_{t \rightarrow \infty} [x \text{sigm}(x - a_i) - s_+(x - a_i)]_0^t \\ &= a_i + s_+(-a_i) = s_+(a_i) \end{aligned}$$

□

Let us now consider two cases:

1. If $f(a_i, 0) > 0$, the basic state is *active*, the unit is generally sensitive and non-zero, but sometimes it is shut off (e.g., when z_i is sufficiently negative to push the argument of the rectifier below 0). In that case gradients will flow in most cases (samples of z_i). If the rest of the system sends the signal that h_i should have been smaller, then gradients will push it towards being more often in the insensitive state.
2. If $f(a_i, 0) = 0$, the basic state is *inactive*, the unit is generally insensitive and zero, but sometimes turned on (e.g., when z_i is sufficiently positive to push the argument of the rectifier above 0). In that case gradients will not flow in most cases, but when they do, the signal will either push the weighted sum lower (if being active was not actually a good thing for that unit in that context) and reduce the chances of being active again, or it will push the weight sum higher (if being active was actually a good thing for that unit in that context) and increase the chances of being active again.

So it appears that even though the gradient does not always flow (as it would with sigmoid or tanh units), it might flow sufficiently often to provide the appropriate training information. The important thing to notice is that even when the basic state (second case, above) is for the unit to be insensitive and zero, *there will be an occasional gradient signal* that can draw it out of there.

One concern with this approach is that one can see an asymmetry between the number of times that a unit with an active state can get a chance to receive a signal telling it to become inactive, versus the number of times that a unit with an inactive state can get a signal telling it to become active.

Another potential and related concern is that some of these units will “die” (become useless) if their basic state is inactive in the vast majority of cases (for example, because their weights pushed them into that regime due to random fluctuations). Because of the above asymmetry, dead units would stay dead for very long before getting a chance of being born again, while live units would sometimes get into the death zone by chance and then get stuck there. What we propose here is a simple mechanism to *adjust the bias of each unit* so that in average its “firing rate” (fraction of the time spent in the active state) reaches some pre-defined target. For example, if the moving average of being non-zero falls below a threshold, the bias is pushed up until that average comes back above the threshold.

2.2 STS Units: Stochastic Times Smooth

We propose here a novel form of stochastic unit that is related to the noisy rectifier and to the stochastic binary unit, but that can be trained by ordinary gradient descent with the gradient obtained by back-propagation in a computational graph. We call it the STS unit (Stochastic Times Smooth). With a_i the activation before the stochastic non-linearity, the output of the STS unit is

$$\begin{aligned} p_i &= \text{sigm}(a_i) \\ b_i &\sim \text{Binomial}(\sqrt{p_i}) \\ h_i &= b_i \sqrt{p_i} \end{aligned} \tag{2}$$

Proposition 2. *The STS unit benefits from the following properties: (a) $E[h_i] = p_i$, (b) $P(h_i > 0) = \sqrt{\text{sigm}(a_i)}$, (c) for any differentiable function f of h_i , we have*

$$E[f(h_i)] = f(p_i) + o(\sqrt{p_i})$$

as $p_i \rightarrow 0$, and where the expectations and probability are over the injected noise b_i , given a_i .

Proof. Statements (a) and (b) are immediately derived from the definitions in Eq. 2 by noting that $E[b_i] = P(b_i > 0) = P(h_i > 0) = \sqrt{p_i}$. Statement (c) is obtained by first writing out the expectation,

$$E[f(h_i)] = \sqrt{p_i} f(\sqrt{p_i}) + (1 - \sqrt{p_i}) f(0)$$

and then performing a Taylor expansion of f around p_i as $p_i \rightarrow 0$:

$$\begin{aligned} f(\sqrt{p_i}) &= f(p_i) + f'(p_i)(\sqrt{p_i} - p_i) + o(\sqrt{p_i} - p_i) \\ f(0) &= f(p_i) + f'(p_i)(-p_i) + o(p_i) \end{aligned}$$

where $\frac{o(x)}{x} \rightarrow 0$ as $x \rightarrow 0$, so we obtain

$$\begin{aligned} E[f(h_i)] &= \sqrt{p_i}(f(p_i) + f'(p_i)(\sqrt{p_i} - p_i)) + (1 - \sqrt{p_i})(f(p_i) + f'(p_i)(-p_i)) + o(\sqrt{p_i}) \\ &= \sqrt{p_i}f(p_i) + f'(p_i)(p_i - p_i\sqrt{p_i}) + f(p_i)(1 - \sqrt{p_i}) - f'(p_i)(p_i - p_i\sqrt{p_i}) + o(\sqrt{p_i}) \\ &= f(p_i) + o(\sqrt{p_i}) \end{aligned}$$

using the fact that $\sqrt{p_i} > p_i$, $o(p_i)$ can be replaced by $o(\sqrt{p_i})$. $\square$

Note that this derivation can be generalized to an expansion around $f(x)$ for any $x \leq \sqrt{p_i}$, yielding

$$E[f(h_i)] = f(x) + (x - p_i)f'(x) + o(\sqrt{p_i})$$

where the effect of the first derivative is canceled out when we choose $x = p_i$. It is also a reasonable choice to expand around p_i because $E[h_i] = p_i$.

3 Unbiased Estimator of Gradient for Stochastic Binary Neurons

The above proposals cannot deal with non-linearities like the indicator function that have a derivative of 0 almost everywhere. Let us consider this case now, where we want some component of our model to take a hard binary decision but allow this decision to be stochastic, with a probability that is a continuous function of some quantities, through parameters that we wish to learn. We will also assume that many such decisions can be taken in parallel with independent noise sources z_i driving the stochastic samples. Without loss of generality, we consider here a set of binary decisions, i.e., the setup corresponds to having a set of stochastic binary neurons, whose output h_i influences an observed future loss L . In the framework of Eq. 1, we could have for example

$$h_i = f(a_i, z_i) = \mathbf{1}_{z_i > \text{sigm}(a_i)} \tag{3}$$

where $z_i \sim U[0, 1]$ is uniform and $\text{sigm}(u) = 1/(1 + \exp(-u))$ is the sigmoid function. We would ideally like to estimate how a change in a_i would impact L in average over the noise sources, so as to be able to propagate this estimated gradients into parameters and inputs of the stochastic neuron.

Theorem 1. *Let h_i be defined as in Eq. 3, with $L = L(h_i, c_i, c_{-i})$ a loss that depends stochastically on h_i , c_i the noise sources that influence a_i , and c_{-i} those that do not influence a_i , then $\hat{g}_i = (h_i - \text{sigm}(a_i)) \times L$ is an unbiased estimator of $g_i = \frac{\partial E_{z_i, c_{-i}}[L|c_i]}{\partial a_i}$ where the expectation is over z_i and c_{-i} , conditioned on the set of noise sources c_i that influence a_i .*

Proof. We will compute the expected value of the estimator and verify that it equals the desired derivative. The set of all noise sources in the system is $\{z_i\} \cup c_i \cup c_{-i}$. We can consider L to be an implicit deterministic function of all the noise sources (z_i, c_i, c_{-i}) , $E_{v_z}[\cdot]$ denotes the expectation over variable v_z , while $E[\cdot|v_z]$ denotes the expectation over all the other random variables besides v_z , i.e., conditioned on v_z .

$$\begin{aligned} E[L|c_i] &= E_{c_{-i}}[E_{z_i}[L(h_i, c_i, c_{-i})]] \\ &= E_{c_{-i}}[E_{z_i}[h_i L(1, c_i, c_{-i}) + (1 - h_i)L(0, c_i, c_{-i})]] \\ &= E_{c_{-i}}[P(h_i = 1|a_i)L(1, c_i, c_{-i}) + P(h_i = 0|a_i)L(0, c_i, c_{-i})] \\ &= E_{c_{-i}}[\text{sigm}(a_i)L(1, c_i, c_{-i}) + (1 - \text{sigm}(a_i))L(0, c_i, c_{-i})] \end{aligned} \quad (4)$$

Since a_i does not influence $P(c_{-i})$, differentiating with respect to a_i gives

$$\begin{aligned} g_i &\stackrel{\text{def}}{=} \frac{\partial E[L|c_i]}{\partial a_i} = E_{c_{-i}}\left[\frac{\partial \text{sigm}(a_i)}{\partial a_i} L(1, c_i, c_{-i}) - \frac{\partial \text{sigm}(a_i)}{\partial a_i} L(0, c_i, c_{-i})|c_i\right] \\ &= E_{c_{-i}}[\text{sigm}(a_i)(1 - \text{sigm}(a_i))(L(1, c_i, c_{-i}) - L(0, c_i, c_{-i}))|c_i] \end{aligned} \quad (5)$$

First consider that since $h_i \in \{0, 1\}$,

$$L(h_i, c_i, c_{-i}) = h_i L(1, c_i, c_{-i}) + (1 - h_i)L(0, c_i, c_{-i})$$

$h_i^2 = h_i$ and $h_i(1 - h_i) = 0$, so

$$\begin{aligned} \hat{g}_i &\stackrel{\text{def}}{=} (h_i - \text{sigm}(a_i))L(h_i, c_i, c_{-i}) = h_i(h_i - \text{sigm}(a_i))L(1, c_i, c_{-i}) \\ &\quad + (h_i - \text{sigm}(a_i))(1 - h_i)L(0, c_i, c_{-i}) \end{aligned} \quad (6)$$

$$= h_i(1 - \text{sigm}(a_i))L(1, c_i, c_{-i}) \quad (7)$$

$$- (1 - h_i)\text{sigm}(a_i)L(0, c_i, c_{-i}). \quad (8)$$

Now let us consider the expected value of the estimator $\hat{g}_i = (h_i - \text{sigm}(a_i))L(h_i, c_i, c_{-i})$.

$$\begin{aligned} llE[\hat{g}_i] &= E[h_i(1 - \text{sigm}(a_i))L(1, c_i, c_{-i}) - (1 - h_i)\text{sigm}(a_i)L(0, c_i, c_{-i})] \\ &= E_{c_i, c_{-i}}[\text{sigm}(a_i)(1 - \text{sigm}(a_i))L(1, c_i, c_{-i}) - (1 - \text{sigm}(a_i))\text{sigm}(a_i)L(0, c_i, c_{-i})] \\ &= E_{c_i, c_{-i}}[\text{sigm}(a_i)(1 - \text{sigm}(a_i))(L(1, c_i, c_{-i}) - L(0, c_i, c_{-i}))] \end{aligned} \quad (9)$$

which is the same as Eq. 5, i.e., the expected value of the estimator equals the gradient of the expected loss, $E[\hat{g}_i] = g_i$. $\square$

This estimator is a special case of the REINFORCE algorithm when the stochastic unit is a Bernoulli with probability given by a sigmoid (Williams, 1992). In the REINFORCE paper, Williams shows that if the stochastic action h is sampled with probability $p_\theta(h)$ and yields a reward R , then

$$E_h[(R - b) \cdot \frac{\partial \log p_\theta(h)}{\partial \theta}] = \frac{\partial E_h[R]}{\partial \theta}$$

where b is an arbitrary constant, i.e., the sampled value h can be seen as a weighted maximum likelihood target for the output distribution $p_\theta(\cdot)$, with the weights $R - b$ proportional to the reward. The additive normalization constant b does not change the expected gradient, but influences its variance, and an optimal choice can be computed, as shown below.

Corollary 1. *Under the same conditions as Theorem 1, and for any (possibly unit-specific) constant $\bar{L}_i$ the centered estimator $(h_i - \text{sigm}(a_i))(L - \bar{L}_i)$, is also an unbiased estimator of $g_i = \frac{\partial E_{z_i, c_{-i}}[L|c_i]}{\partial a_i}$. Furthermore, among all possible values of $\bar{L}_i$, the minimum variance choice is*

$$\bar{L}_i = \frac{E[(h_i - \text{sigm}(a_i))^2 L]}{E[(h_i - \text{sigm}(a_i))^2]}, \quad (10)$$

which we note is a weighted average of the loss values L , whose weights are specific to unit i .

Proof. The centered estimator $(h_i - \text{sigm}(a_i))(L - \bar{L}_i)$ can be decomposed into the sum of the uncentered estimator $\hat{g}_i$ and the term $(h_i - \text{sigm}(a_i))\bar{L}_i$. Since $E_{z_i}[h_i|a_i] = \text{sigm}(a_i)$, $E[\bar{L}_i(h_i - \text{sigm}(a_i))|a_i] = 0$, so that the expected value of the centered estimator equals the expected value